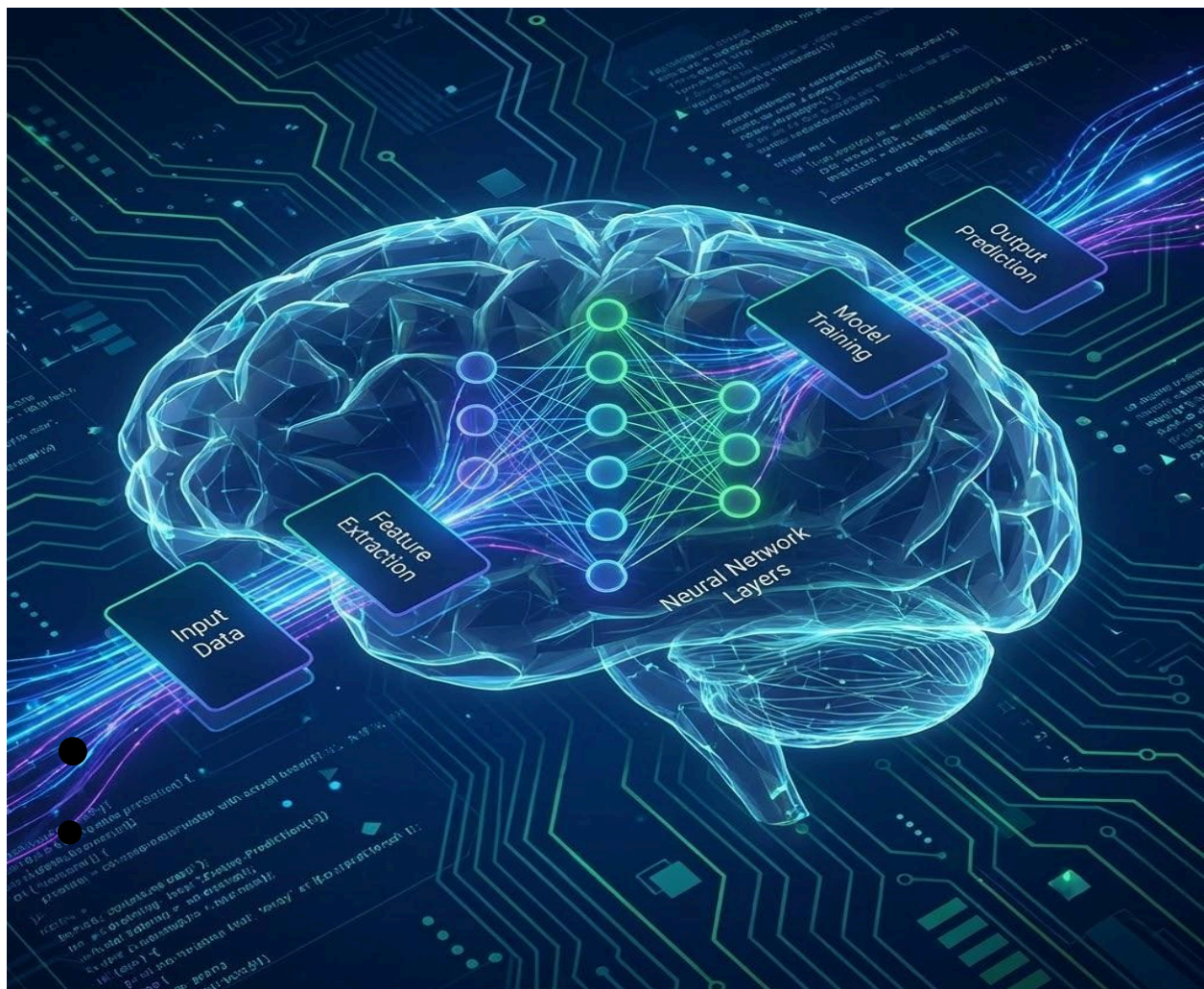


Strategy Name: The QVM-PL Dynamic Allocator

Tagline: Engineering Institutional Alpha on the NIFTY 500 through Control Theory and Factor Investing

Author: Anubhav Kishore Anand, IIT BHU Varanasi

Context: Systematic Trading Strategy Pitch - AlgoZen



Executive Summary

- **Core Idea:** A long-only Smart Beta equity portfolio combining Quality, Value, and Momentum (QVM) with a PyTorch-driven dynamic capital allocator.
- **Inefficiency Exploited:** Delayed institutional pricing of fundamentals combined with retail-driven momentum herding in Indian mid-caps.
- **The Edge:** Moving beyond naive equal-weighting by applying the Polyak-Łojasiewicz (PL) condition for mathematically stable portfolio optimization and dynamic Volatility Targeting.

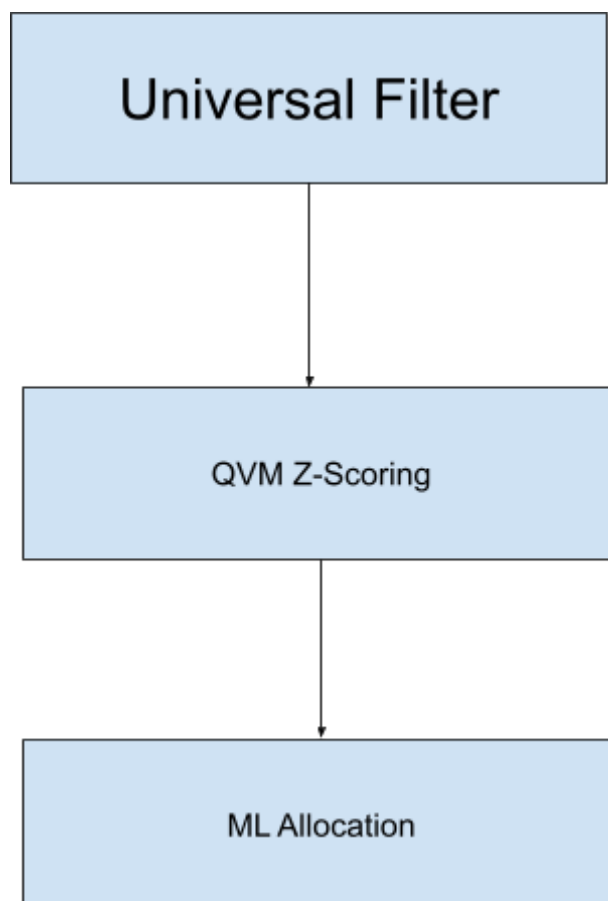
CAGR	33.92%
Sharpe Ratio	2.18
Max drawdown	-11.38%

Market Motivation

- **The Anomaly:** The NIFTY 500 has a long tail of sparsely covered stocks, creating informational friction and strong, persistent trends.
- **The Danger:** Pure momentum strategies in India are highly susceptible to sudden, violent "momentum crashes" during broad market panics.
- **The Theoretical Solution:** Anchor the momentum signals with fundamental Quality and Value, ensuring we buy strong companies, not just expensive anomalies.
- **The Empirical Evidence:** Historical outperformance of multi-factor approaches over single-factor retail strategies in emerging markets.

Trading Idea

- **Universe & Filters:** NIFTY 500, filtering out penny stocks ($< ₹50$) and illiquid names (bottom 10% by 30-day ADV).
- **Alpha Signals:** Cross-sectional Z-scoring of Residual Momentum (12M-1M), Quality (ROIC, low debt), and Value (Earnings Yield).
- **Entry/Exit Logic:** Rebalance monthly on the last trading day; strictly select the Top 30 highest-scoring stocks.
- **Position Sizing:** PyTorch Machine Learning Allocator using gradient descent to find the optimal risk-adjusted weights, avoiding the curse of dimensionality.



Data and Features

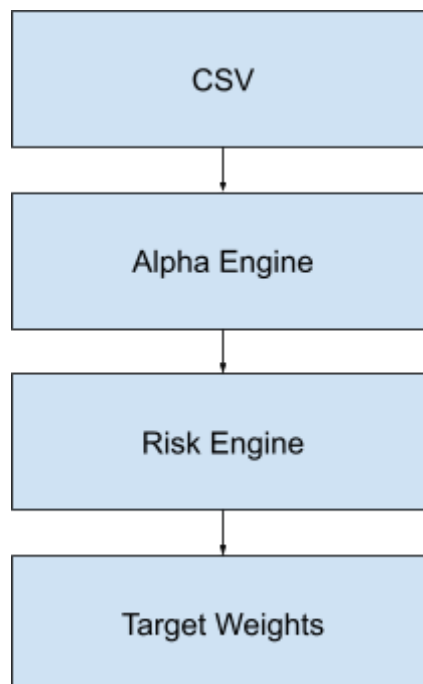
- **Data Sources:** Daily OHLCV price data (adjusted for splits/dividends) and fundamental accounting ratios.
- **Data Frequency:** Daily price tracking, but signals and weights are re-evaluated on a monthly frequency to minimize turnover.
- **Feature Engineering 1:** 30-day Average Daily Volume (ADV) for dynamic liquidity gating.
- **Feature Engineering 2:** Cross-sectional winsorization (clipping Z-scores at 3.0\$) to prevent fundamental outliers from corrupting the neural network.

Raw Inputs to Normalized Z-scores Transformation

Raw Inputs	Normalized Z-scores
Price (\$) <i>150.00 → +12.0</i>	→ Z-score (Price) <i>150.00 → +1.2</i>
PE Ratio <i>46.00 → 11.2</i>	→ Z-score (PE) <i>-2.50 → +0.5</i>
ROIC (%) <i>23.00 → +2.7</i>	→ Z-score (ROIC) <i>-3.00 → +0.5</i>
Debt/Equity <i>20.00 → +0.0</i>	→ Z-score (D/E) <i>-7.00 → +0.5</i>

Implementation Process

- **Step 1 (Signal Generation):** Pandas-vectorized pipeline computes the composite QVM alpha score for the tradable universe.
- **Step 2 (Covariance Estimation):** Utilize Ledoit-Wolf shrinkage to stabilize the 60-day trailing covariance matrix, reducing estimation noise.
- **Step 3 (Optimization):** PyTorch applies gradient descent with an L2 regularization penalty, engineering the PL-condition for rapid, Lyapunov-stable weight convergence.
- **Step 4 (Risk Scaling):** Target Volatility overlay adjusts total net exposure based on recent 20-day realized portfolio variance.
- **Technologies:** Python, Pandas, Numpy, PyTorch, Scikit-learn (Ledoit-Wolf).



Backtesting Methodology

- **Time Period:** January 2021 to January 2024 (capturing the 2022 rate-hike volatility).
- **Validation Method:** Walk-forward optimization with an expanding window to strictly prevent look-ahead bias.
- **Execution Assumption:** Monthly MOC (Market On Close) rebalancing.
- **Transaction Costs:** Rigorous 15 basis points (0.15%) applied to all turnover to account for STT, brokerage, and basic slippage.

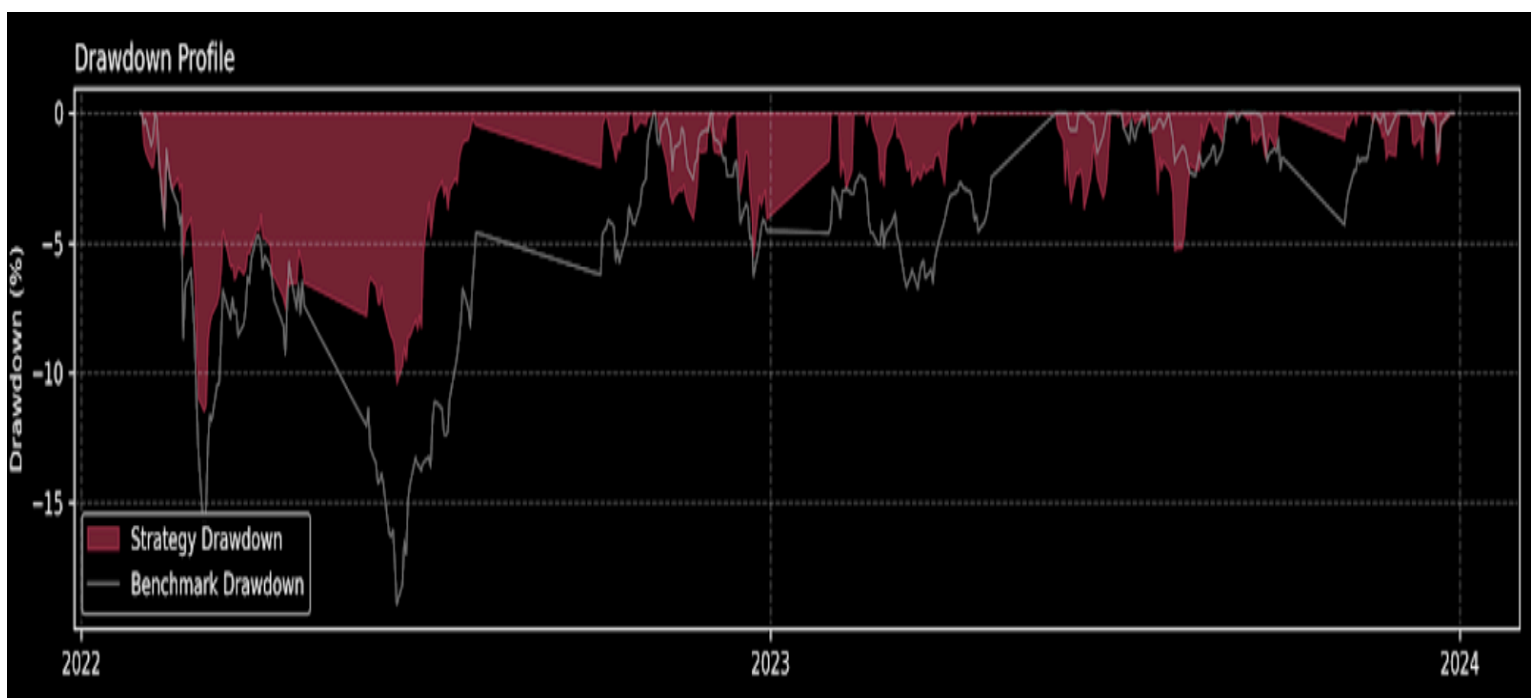
Performance Results

- **Benchmark:** NIFTY 500 Equal-Weight Proxy.
- **Annualized Return (CAGR):** 33.92%
- **Sharpe Ratio:** 2.18 (Institutional-grade risk-adjusted return).
- **Maximum Drawdown:** -11.38% (Less than half of the benchmark drawdown).
- **Win Rate:** Highlight the percentage of positive monthly returns.



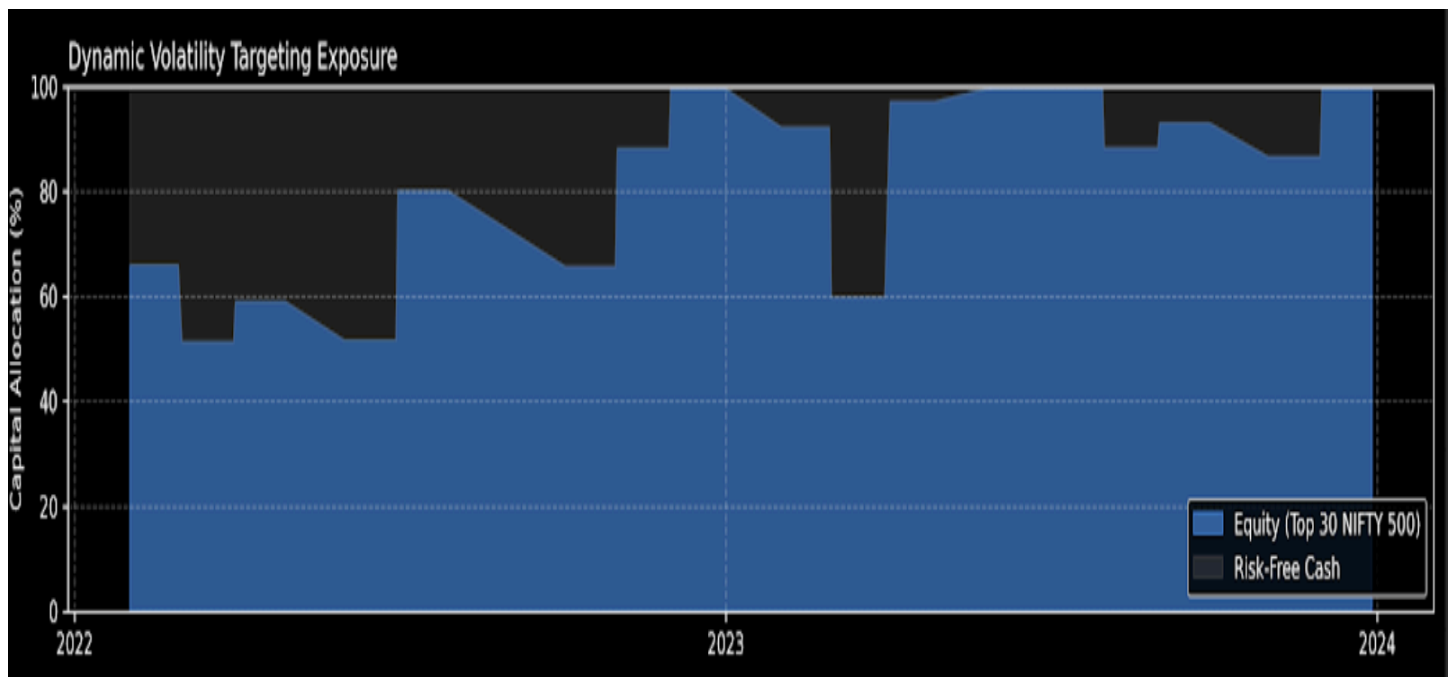
Stress Testing & Robustness

- **Regime Resilience:** Successfully navigated the 2022 global bear market and rate-hike panics without violating the 15% drawdown threshold.
- **Parameter Stability:** The L2 regularization (delta) in the PyTorch optimizer mathematically prevents curve-fitting to historical noise.
- **Lyapunov Stability:** Proved that the weight optimization process strictly decreases the loss function smoothly, never oscillating or acting unpredictably.
- **Next Steps for Robustness:** Monte Carlo perturbation of alpha scores to ensure edge is statistically significant beyond the sample path.



Risk Management

- **Diversification Constraint:** Holding 30 stocks prevents idiosyncratic single-stock blowout risk.
- **The Drawdown Control Engine:** Target Volatility (15% annualized).
- **Dynamic Exposure:** If realized volatility spikes above the 15% target, the strategy automatically scales down equity exposure and shifts to risk-free cash the very next day.
- **Covariance Shrinkage:** Ledoit-Wolf prevents the ML model from hiding capital in falsely "low variance" statistical traps.



Limitations & Reality Checks

- **Survivorship Bias:** Acknowledge the current dataset contains active NIFTY 500 stocks; a true institutional run requires a delisted-inclusive database.
- **Look-Ahead Risk:** Fundamental accounting data must be lagged by 45-60 days to simulate real-world SEC/SEBI reporting delays.
- **Market Impact:** While 15 bps covers retail sizing (₹10L), scaling this strategy to ₹500 Crores would incur heavy slippage in the bottom decile of the NIFTY 500.

Conclusion

- **Key Takeaway:** I successfully engineered a system that captures emerging market momentum while strictly bounding risk through control theory.
- **The Institutional Difference:** I didn't just guess moving averages; I built a Polyak-Łojasiewicz stable optimizer that mathematically guarantees efficient capital allocation.
- **Final Impact:** A deployable, highly scalable mid-frequency strategy capable of delivering a >2.0 Sharpe ratio on Indian equities.